

Equations

Formulas and Equations

2026

Preliminary Mathematics Standard

Equations

Homework

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What is a Linear Equation?

Definition

A **linear equation** has:

- Variables with exponent 1 only
- No products of variables
- Constant coefficients

Example

$$ax + b = 0$$

where $a \neq 0$

Examples

- $3x + 5 = 17$
- $2x - 7 = x + 3$
- $\frac{x}{4} + 2 = 10$
- $5(2x - 1) = 3x + 12$

Non-Examples

- $x^2 + 3x = 5$ (quadratic)
- $xy = 12$ (product of variables)
- $\sqrt{x} = 4$ (not linear)

Translating Words to Mathematics

Words	Math Symbol
is, equals, gives	=
sum, total, plus, increased by	+
difference, minus, decreased by	-
product, times, multiplied by	$\times$ or $\cdot$
quotient, divided by, per	$\div$ or fraction
more than	+
less than	-
twice, double	$2\times$
half, one-half	$\frac{1}{2}\times$

Example

- "5 more than x ": $x + 5$
- "Twice a number": $2x$
- "3 less than y ": $y - 3$
- "Half of n ": $\frac{n}{2}$

Watch Out!

- "5 less than x " = $x - 5$ NOT $5 - x$
- Order matters!

5-Step Problem Solving Process

1. Read & Understand

- What are you asked to find?
- What information is given?

2. Define Variables

- Let $x = \dots$
- Label clearly

3. Write Equation

- Translate words to algebra
- Use key words table

4. Solve Equation

- Use algebraic methods
- Show all working

5. Interpret Solution

- Answer in context
- Check reasonableness

Example 1: Simple Number Problem

Example

Three more than twice a number is 17. Find the number.

Step 1: Understand

We need to find an unknown number.

Step 2: Define Variable

Let x = the number

Step 3: Write Equation

- "Twice a number": $2x$
- "Three more than": $+3$
- "is 17": $= 17$

$$2x + 3 = 17$$

Example 1: Solving & Interpreting

Step 4: Solve Equation

$$2x + 3 = 17$$

$$2x + 3 - 3 = 17 - 3$$

$$2x = 14$$

$$\frac{2x}{2} = \frac{14}{2}$$

$$x = 7$$

Step 5: Interpret Solution

- The number is 7
- Check: Twice 7 is 14, plus 3 is 17
- This is a reasonable solution

Key Insight

The equation $2x + 3 = 17$ represents the word problem exactly!

Example 2: Age Problem

Example

Jane is 5 years older than Tom. In 3 years, the sum of their ages will be 31. Find their current ages.

Define Variables

Let:

- T = Tom's current age
- J = Jane's current age

Given Information

- Now: $J = T + 5$
- In 3 years:
 - Tom: $T + 3$
 - Jane: $J + 3$
- Sum = 31

Example 2: Solving

Write Equation

$$(T + 3) + (J + 3) = 31$$

But $J = T + 5$, so:

$$(T + 3) + ((T + 5) + 3) = 31$$

Simplify:

$$(T + 3) + (T + 8) = 31$$

$$2T + 11 = 31$$

Solve

$$2T + 11 = 31$$

$$2T = 20$$

$$T = 10$$

$$J = T + 5 = 10 + 5 = 15$$

Example 2: Interpretation & Check

Solution

- Tom is 10 years old
- Jane is 15 years old

Check

- Jane is 5 years older than Tom: $15 - 10 = 5$
- In 3 years:
 - Tom: $10 + 3 = 13$
 - Jane: $15 + 3 = 18$
 - Sum: $13 + 18 = 31$

Important

- Age problems often involve "now" and "future/past"
- Define ages carefully
- Check in both time frames

Example 3: Measurement Problem

Example

A rectangle has length 3 cm more than its width. If the perimeter is 26 cm, find the dimensions.

Step 1 & 2

- Need: width and length
- Let w = width (cm)
- Length: $w + 3$ (cm)

Perimeter Formula

$$P = 2 \times \text{length} + 2 \times \text{width}$$

$$P = 2l + 2w$$

Step 3: Write Equation

$$2(w + 3) + 2w = 26$$

Example 3: Solving

Solve Equation

$$2(w + 3) + 2w = 26$$

$$2w + 6 + 2w = 26$$

$$4w + 6 = 26$$

$$4w = 20$$

$$w = 5$$

Find Length

$$l = w + 3 = 5 + 3 = 8$$

Interpret

- Width = 5 cm
- Length = 8 cm
- Check: Perimeter = $2(8) + 2(5) = 16 + 10 = 26$
- Reasonable dimensions

Common Mistakes to Avoid

Mistake #1

- **What:** Wrong variable order
- **Example:** "5 less than x "
 - Correct: $x - 5$
 - Wrong: $5 - x$

Mistake #2

- **What:** Forgetting units
- **Fix:** Always include units in answer

Mistake #3

- **What:** No interpretation
- **Fix:** Always answer in context

Mistake #4

- **What:** Not checking reasonableness
- **Example:** Negative age, negative length
- **Fix:** Ask "Does this make sense?"

Your Turn: Practice Problem

Example

A taxi charges \$3.50 flag fall plus \$2.10 per kilometre. If the fare is \$25.70, how far did the taxi travel?

Think About:

1. What are you asked to find?
2. What information is given?
3. What variable should you use?
4. What equation represents this situation?

$$\text{The total cost} = \text{fixed cost} + (\text{rate} \times \text{distance})$$

Practice Problem Solution

Step 1 & 2

- Find: distance travelled (km)
- Let d = distance (km)

Step 3: Equation

$$3.50 + 2.10d = 25.70$$

Step 4: Solve

$$2.10d = 25.70 - 3.50$$

$$2.10d = 22.20$$

$$d = \frac{22.20}{2.10}$$

$$d = 10.57 \dots \approx 10.6 \text{ km}$$

Example

Challenge Question "The sum of three consecutive even numbers is 78.
Find the numbers."

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Complete all questions from *Formulas and Equations — Equations*.